

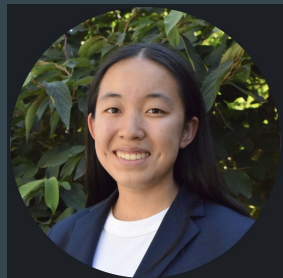
Equitable Park Sector Optimization



DecisionLab 2026

Xinying Bi, Vincent Chang, Selina Gu, Sevyn Liu, Jeanne Manteau

Team — DecisionLab 2026



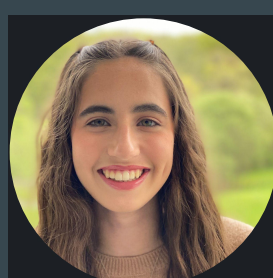
Xinying Bi



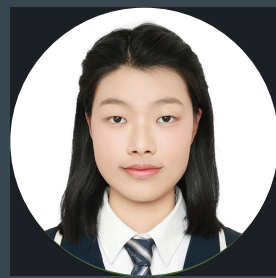
Vincent Chang



Sevyn Liu



Jeanne Manteau



Selina Gu

Community Partners

Ed Pettitt — Friends of Columbia Tap

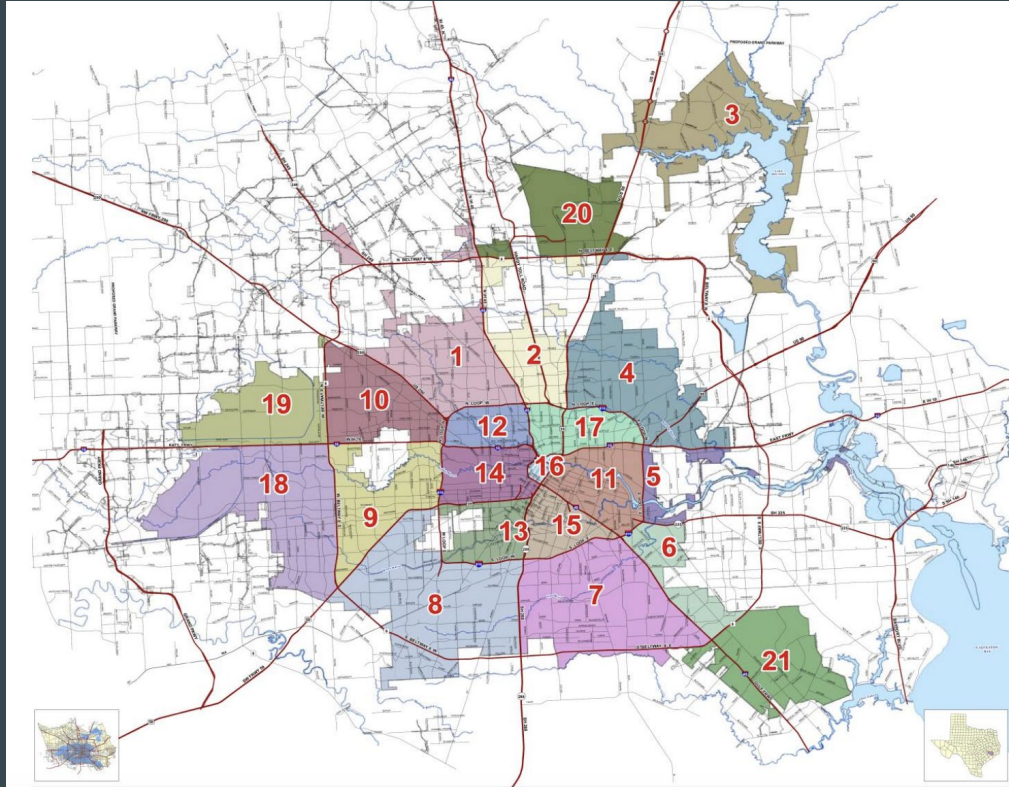
Taylor Valley — Coalition for Environment,
Equity, and Resilience (CEER)

Mentors

Samuel Garvin (Optimization Mentor)

Snikitha Kassey (Development Mentor)

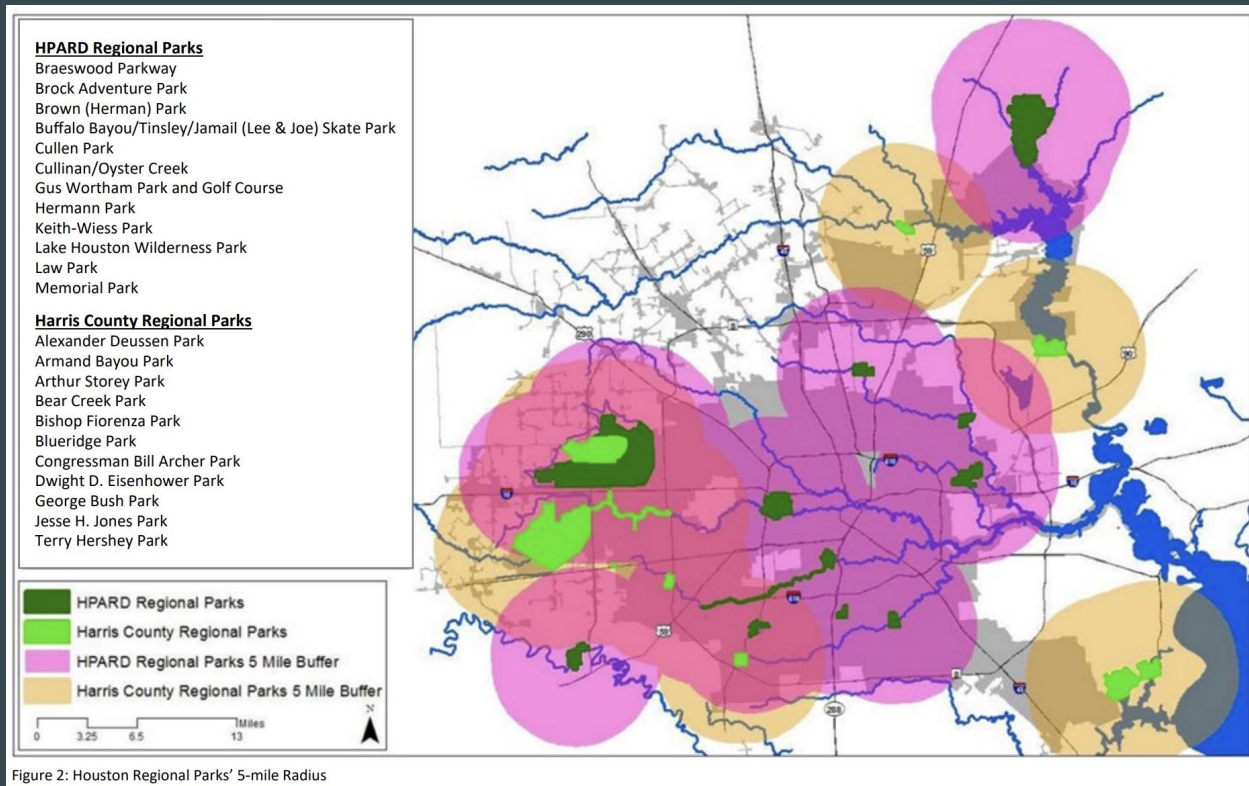
Introduction - Houston's Current Park Sector System



Current System:

- 21 park sectors
- Boundaries follow highways
- Fees stay within sector
- Not based on need

Why Current Sectors Aren't Equitable & Our Goal



The Problem:

- Funding imbalance
- Low park access
- Poor quality
- High heat / low trees

Our Goal:

- Align funding with need
- Improve equity
- Respect real constraints

Data - Plats

- Houston Park development is funded by newly developed “dwelling units”
- Developers have a choice of either dedicating land or paying fees in lieu of land dedication to HPARD.
- All of the funding HPARD can use comes from these fees in lieu.

Data - Plats



- We only have data for the amount of fees in lieu for each park sector from years 2008-2021
- We must get data of fees in lieu for smaller sections than park sectors for more recent years
- Choose census tracts for our sectors

App No. ↓	Subdivision Name	App Type	Application Status	Date Submitted	Applicant Name	Council District	Super Neighborhood	County Precinct	Agenda	
2026-0541	Jubilee Commercial Reserve	C2	Updating Action Form	4/6/2026	Kenya Guajardo			H4	2026-04-16	✔
2026-0540	Bridgeland Prairieland Village Sec 56	C3F	Updating Action Form	4/6/2026	Bob Terry			H4	2026-04-16	✔
2026-0537	Huffman Blue Lake	C2	Updating Action Form	4/6/2026	Alan Torres			H3	2026-04-16	✔
2026-0536	Wycliffe Manor	C2R	Updating Action Form	4/6/2026	Tom Duecker	A	SPRING BRANCH WEST	H4	2026-04-16	✔
2026-0535	Winward Falls GP	GP	Action Form Completed	4/6/2026	Jacob Guerrero			H4	2026-04-16	✔
2026-0534	Becker Road Tract GP	GP	Action Form Completed	4/6/2026	Jacob Guerrero			H4	2026-04-16	✔
2026-0532	La Branch Square	C2R	Updating Action Form	4/6/2026	Anup Pande	D	MUSEUM PARK	H1	2026-04-16	✔
2026-0531	Diamond Reserve	C3F	Updating Action Form	4/6/2026	Karla Martinez	C	GREATER HEIGHTS	H1	2026-04-16	✔
2026-0528	Falcon Express at West Mount	C2	Updating Action Form	4/6/2026	Shahnawaz Ghanchi			H2	2026-04-16	✔
2026-0526	Hempstead Jackrabbit	C2	Updating Action Form	4/6/2026	Alan Torres			H3	2026-04-16	✔

Land Use *

Single Family Residential (Shared Driveways)

Subdivision

Plat Data

Location & Contacts

Documents

Other Submissions

Park and Open Space (If applicable)

Number of Existing Dwelling Units *

12

Number of Proposed Dwelling Units *

10

Total Number of Units

Park Fees *

☐ No ☒ Yes

Private Deduction in Acreages *

0.0000

Public Dedication in Acreages *

0.0000

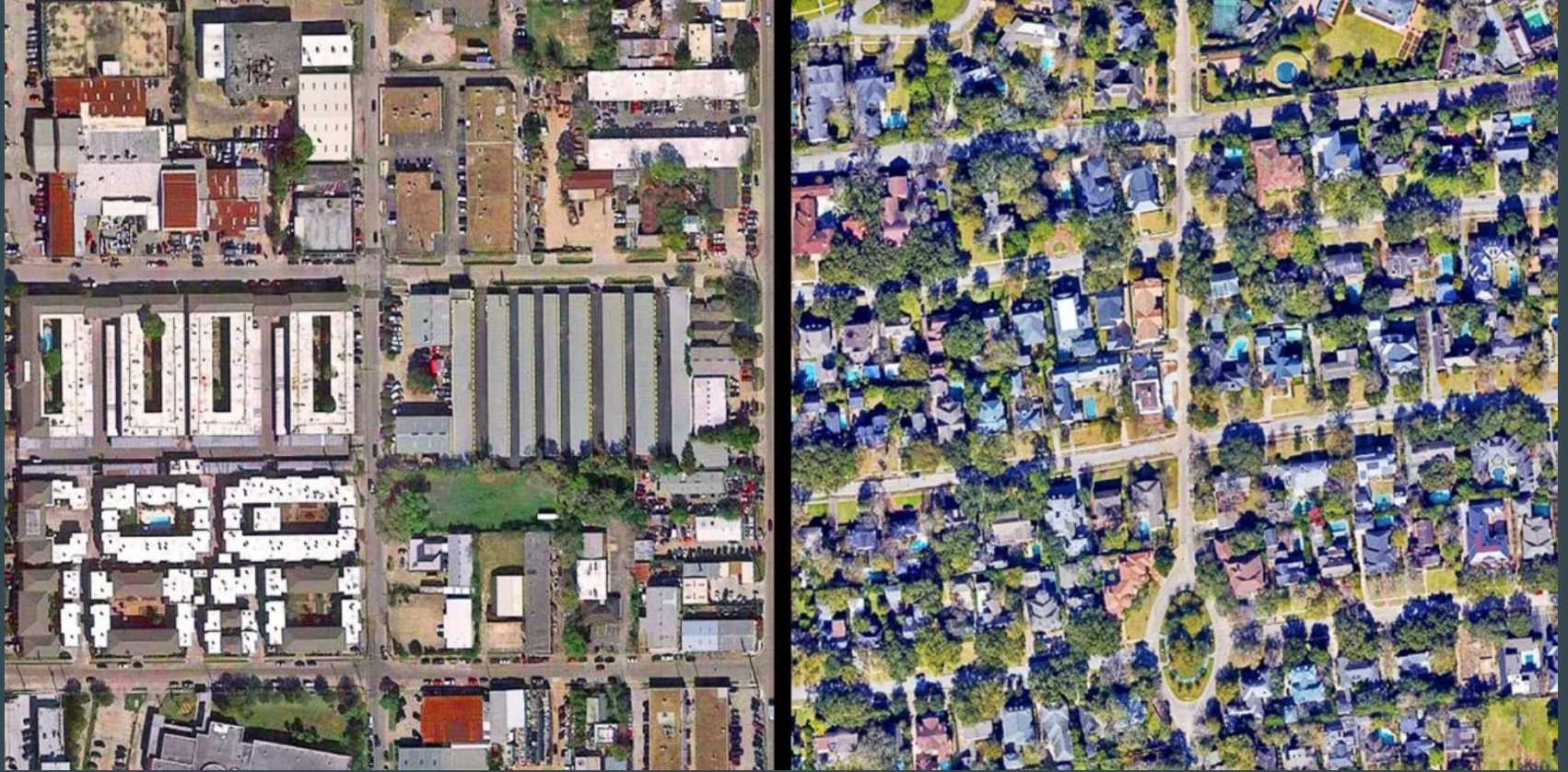
Proposed Street Names

App No.	Street Prefix	Street Name ↑	Street Type	Street Suffix	Street Category	Status	Addressing Final Status	Planner Status

Data - Plats

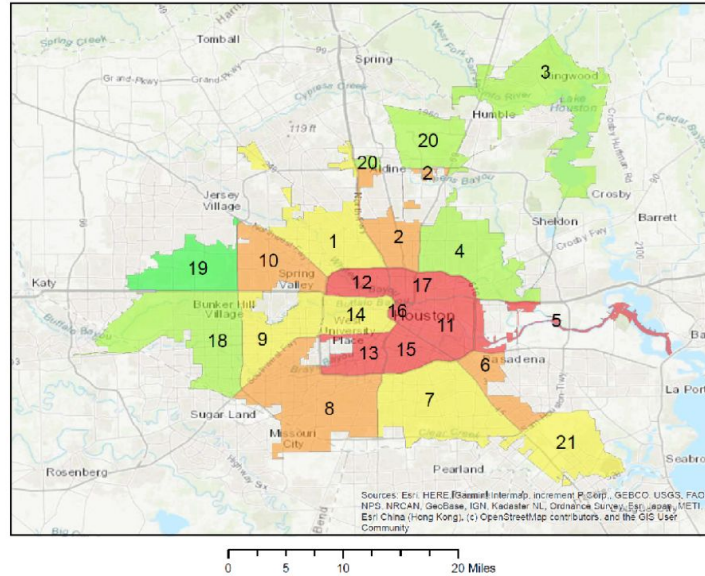
- After scraping all submissions for years 2024 and 2025, we get 10,296 units of fees in lieu and 10,581 respectively
- This gives us \$7,207,200 and \$7,406,700 of funding for years 2024 and 2025 respectively
- Compared their average our most recent data of funding (lump sum of 2022-Summer 2025), we underestimate the true value by 13%

Data - Greenspace

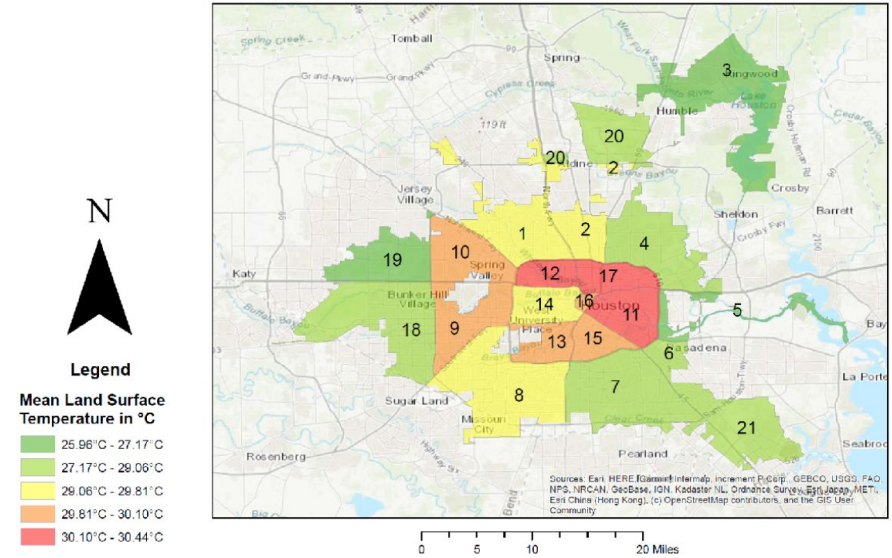


Data - Greenspace

Percent Tree Canopy by Park Sector in Houston, Texas



Mean Land Surface Temperature by Park Sector in Houston, Texas



HarrisTreeCanopy

MEAN

◀ > 23.000000

◀ 14.100000

◀ < 5.000000

HarrisTreeCanopy



MEAN



◀ > 23.000000

◀ 14.100000

◀ < 5.000000

HarrisTreeCanopy



MEAN



◀ > 23.000000

◀ 14.100000

◀ < 5.000000

HarrisTreeCanopy



MEAN



◀ > 23.000000

◀ 14.100000

◀ < 5.000000

HarrisTreeCanopy



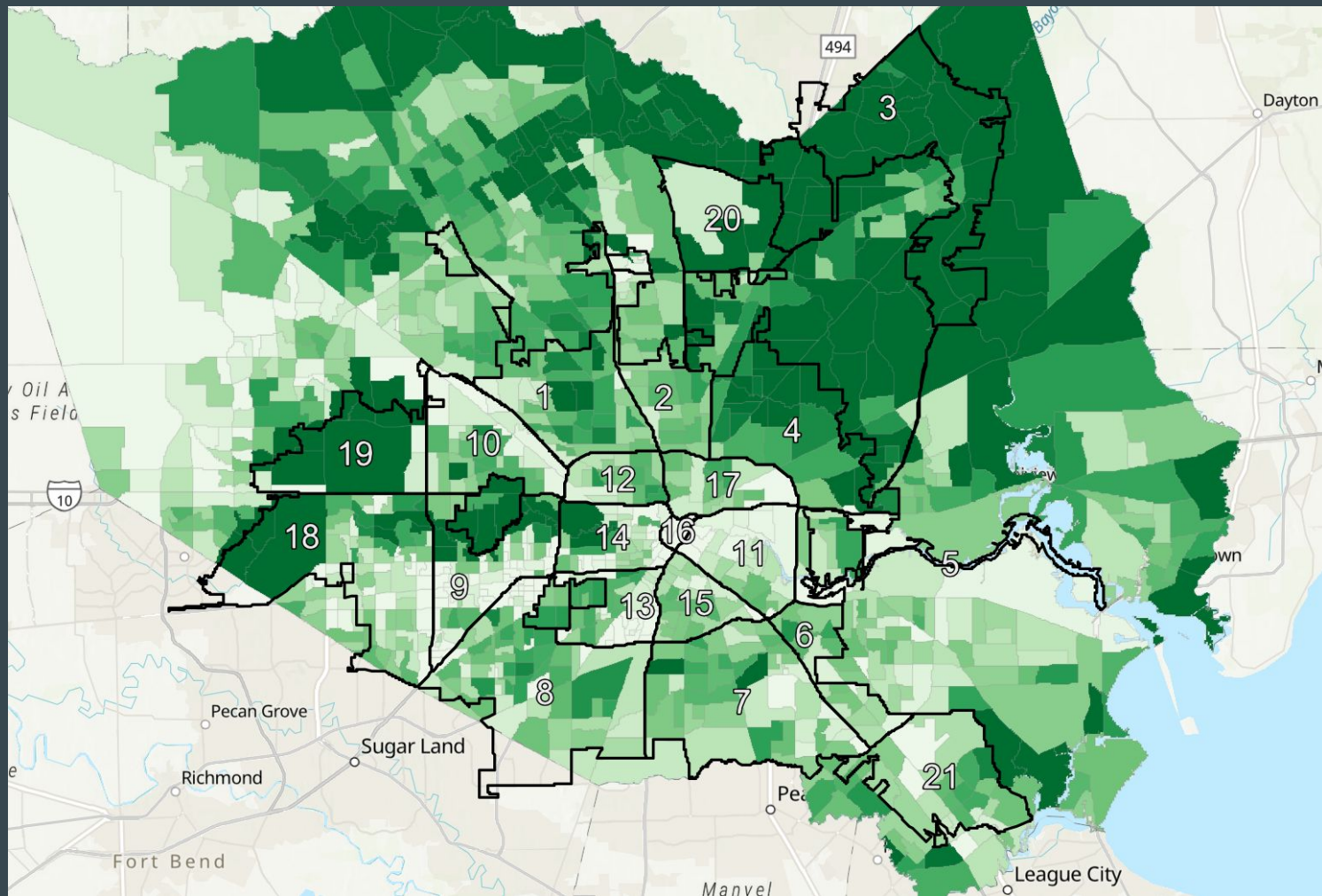
MEAN



◀ > 23.000000

◀ 14.100000

◀ < 5.000000



Model Formulation

Sets

D = set of census tracts

N = set of sectors, $|N| = 21$ (currently)

Decision Variables

$$x_{ij} = \begin{cases} 1 & \text{if census tract } i \text{ is assigned to sector } j \\ 0 & \text{otherwise} \end{cases}$$

ε_j = revenue imbalance of sector j

Parameters

c_i = fees generated in census tract i

$$\theta = \frac{\sum_{i \in D} c_i}{|N|} \quad \text{ideal average revenue per sector}$$

Objective

$$\min \sum_{j \in N} \varepsilon_j$$

Subject to

$$\sum_{j \in N} x_{ij} = 1 \quad \forall i \in D \quad (\text{each census tract assigned to one sector})$$

$$\left| \sum_{i \in D} c_i x_{ij} - \theta \right| \leq \varepsilon_j \quad \forall j \in N \quad (\text{sector revenue close to ideal average})$$

$$x_{ij} \in \{0, 1\} \quad \forall i \in D, j \in N$$

$$\varepsilon_j \geq 0 \quad \forall j \in N$$

Model Formulation

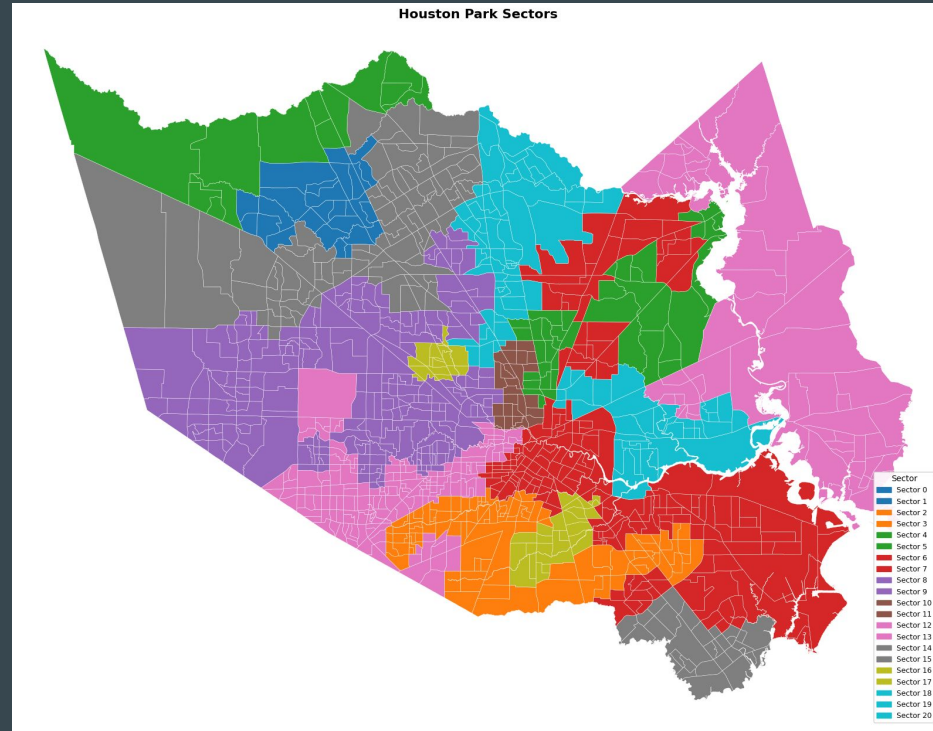
Contiguity constraint:

- for each sector, we choose one tract as the “root”
- every other tract assigned to that sector must be reachable from the root

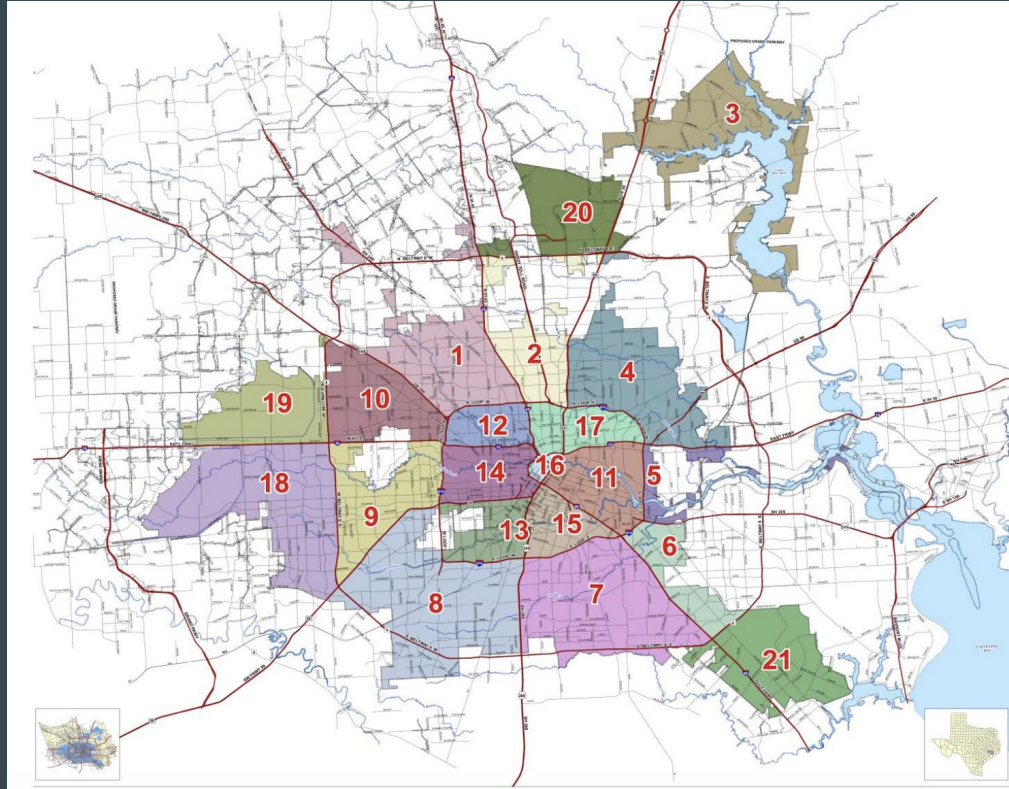
Compactness constraint:

- we don't want one sector stretching across the city in a long, thin shape
- the distance from the district to the center of the sector should be smaller than a radius R

Results



Results



Old Sectors

PARK SECTORS	# OF UNITS
1	631
2	53
3	43
4	137
5	4
6	57
7	529
8	373
9	55
10	358
11	287
12	1,571
13	389
14	1,293
15	143
16	0
17	99
18	325
19	448
20	2
21	2

Total
Imbalance:
6006

New Formulation

PARK SECTORS	# OF UNITS	Greenspace
1	0	460
2	0	0
3	400	274
4	249	808
5	272	837
6	206	287
7	561	851
8	269	1690
9	282	1680
10	249	1441
11	0	0
12	278	348
13	208	1395
14	268	1550
15	311	1494
16	0	390
17	513	159
18	564	336
19	1016	1225
20	275	589
21	0	0

Total
Imbalance:
3346

Limitations and Future Plans

- Equal funding per sector may not be most ideal
 - Did not take into account existing infrastructure
 - Did not consider population characteristics in sectors
 - 70/30 Plan
- Find more recent and encompassing information
- Incorporate existing infrastructure (parks, playgrounds, greenhouses, etc.)

Thank You!

Questions?